

Week 1: System Planning & Distribution Selection

System Architecture Description:

Host Machine (Windows 11):

- Runs VirtualBox
- Provides NAT internet connectivity for both VMs
- Provides Host-Only virtual network (vboxnet0)

Workstation VM (Linux Mint 21.3)

- Purpose: Administrative machine
- Used to SSH into the server and perform management
- Runs GUI for convenience
- IP (Host-Only): 192.168.56.11

Server VM (Ubuntu Server 24.04 LTS)

- Purpose: Headless Linux server
- All configuration performed exclusively via SSH
- No desktop environment
- IP (Host-Only): 192.168.56.10

Network Connections:

- Adapter 1 (NAT): Internet access to install updates
- Adapter 2 (Host-Only): Internal isolated network used for SSH and monitoring
- Ensures secure, controlled communication between workstation and server only

Distribution Selection Justification:

Chosen Server OS: Ubuntu Server 24.04 LTS

Ubuntu Server is selected because:

- **Long-term support (5 years):** Ideal for secure deployments without frequent upgrades
- **Industry standard for cloud and DevOps:** Widely used in AWS, Azure, GCP
- **Large package repository & documentation:** Easier troubleshooting and availability of security tools
- **Headless-friendly:** Designed for remote CLI management and SSH administration

- **Strong hardware compatibility:** Supports modern virtualization environments such as VirtualBox and VMware

Workstation Distribution Linux Mint 21.3

Linux Mint is chosen as the workstation for these reasons;

- **User-friendly GUI:** Perfect for taking screenshots, documentation, and running commands easily
- **Ubuntu-based:** Works seamlessly with Ubuntu Server due to shared package ecosystem
- **Lightweight and responsive:** Good performance even with limited resources
- **SSH client available out-of-the-box:** Makes it easy to connect to the Ubuntu Server

Workstation Configuration Decision

The option I went with is the Linux Desktop VM as my workstation environment. This workstation runs Linux Mint 21.3, which provides a lightweight GUI, a built-in terminal, and full compatibility with Ubuntu Server since both share the same Ubuntu package ecosystem

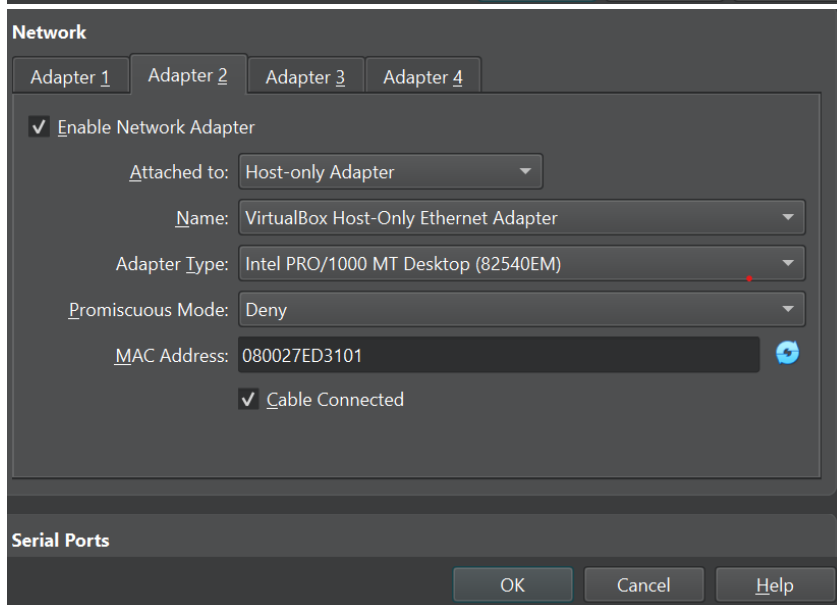
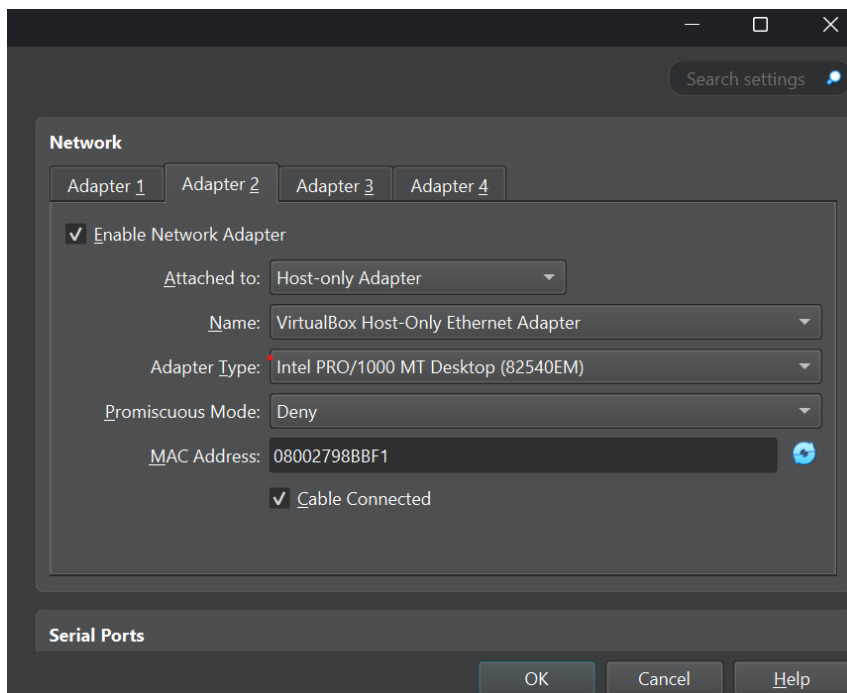
Using a Linux VM as the workstation offers several advantages:

- **Full isolation:** All administrative tasks are performed inside a controlled environment, preventing conflicts with host operating system.
- **Consistency and reliability:** A Linux workstation ensures consistent behaviour with the Ubuntu Server commands, tools, and package managers used throughout the coursework
- **Built-in SSH Client:** Mint includes OpenSSH and terminal tools by default, allowing seamless SSH access to the server VM

Compared to the other alternatives, the Linux desktop VM is better since the Host Machine risks network configuration conflicts and hybrid adds unnecessary complexity and increases the chance of misconfiguration

Network Configuration Documentation

In this section, I created a Host-Only network for both the VMs. Below attached are screenshots showing the network settings



The first one is for the Ubuntu server and the other is for the Linux Mint workstation

The next thing I did was set a static IP on the Ubuntu Server. I did this by opening the terminal in the Ubuntu server. Then after that I typed up the `ip addr` command to find the `enp0s3` and `enp0s8`. After that, I edited the netplan config using `sudo nano /etc/netplan/01-netcfg.yaml`. I replaced the `enp0s8` with my host only interface by adding this to the .yaml file

```

network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s3:
      dhcp4: true      # NAT adapter
    enp0s8:
      addresses:
        - 192.168.56.10/24
      routes:
        - to: 0.0.0.0/0
          via: 192.168.56.1
      nameservers:
        addresses: [8.8.8.8, 1.1.1.1]

```

Then applying it by running `sudo netplan apply`

Below attached is evidence that the change has happened:

```

3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP
group default qlen 1000
    link/ether 08:00:27:98:bb:f1 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.10/24 brd 192.168.56.255 scope global noprefixroute enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe98:bbf1/64 scope link
        valid_lft forever preferred_lft forever

```

I tested the network by pingping google.com. This was successful and the results were:

Ubuntu: 10 packets received, 0% packet loss, time 13435ms

rtt min/avg/max/mdev = 16.768/23.765/37.293/6.134

Mint: 10 packets received, 0% packets loss, time 9387ms

rtt min/avg/max/mdev = 14.345/29.334/120.294/31.441

I also ran a bunch of terminal commands in both Ubuntu and Mint:

```
vboxuser@Ubuntu:~$ uname -a
Linux Ubuntu 6.8.0-49-generic #49-Ubuntu SMP PREEMPT_DYNAMIC Mon Nov 4 02:06:24
UTC 2024 x86_64 x86_64 GNU/Linux
vboxuser@Ubuntu:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           7.8Gi         1.0Gi         6.0Gi          32Mi         1.0Gi         6.7Gi
Swap:            0B           0B           0B
vboxuser@Ubuntu:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           795M  1.5M  793M   1% /run
/dev/sda2       25G   7.3G   16G  32% /
tmpfs           3.9G   0     3.9G   0% /dev/shm
tmpfs           5.0M  8.0K   5.0M   1% /run/lock
tmpfs           795M  120K  795M   1% /run/user/1000
vboxuser@Ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:e4:d9:e7 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 85651sec preferred_lft 85651sec
    inet6 fd00::cc24:fee:3f45:ce05/64 scope global temporary dynamic
        valid_lft 85901sec preferred_lft 13901sec
    inet6 fd00::a00:27ff:fee4:d9e7/64 scope global dynamic mngtmpaddr
        valid_lft 85901sec preferred_lft 13901sec
    inet6 fe80::a00:27ff:fee4:d9e7/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:98:bb:f1 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.10/24 brd 192.168.56.255 scope global noprefixroute enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe98:bbf1/64 scope link
        valid_lft forever preferred_lft forever
vboxuser@Ubuntu:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.1 LTS
Release:        24.04
Codename:       noble
vboxuser@Ubuntu:~$
```

```
mint@mint:~$ uname -a
Linux Mint 6.14.0-29-generic #29-24.04.1-Ubuntu SMP PREEMPT_DYNAMIC Thu Aug 14 16:52:50 UTC 2 x86_64 x86_64 x86_64 GNU/Linux
mint@mint:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.6Gi         932Mi         180Mi          180Mi         3.0Gi         2.9Gi
Swap:            0B           0B           0B
mint@mint:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           392M  1.2M  391M   1% /run
dev/sr0         2.9G  2.9G   0 100% /cdrom
cow             2.0G  178M   1.8G  10% /
tmpfs           2.0G   0     2.0G   0% /dev/shm
tmpfs           5.0M  8.0K   5.0M   1% /run/lock
tmpfs           2.0G  8.0K   2.0G   1% /tmp
tmpfs           392M  156K  392M   1% /run/user/1000
mint@mint:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:3c:20:15 brd ff:ff:ff:ff:ff:ff
    link/ether 08:00:27:3c:20:15 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 80357sec preferred_lft 80357sec
    inet6 fd00::7010:b977:96a1:2b4e/64 scope global temporary dynamic
        valid_lft 86359sec preferred_lft 14359sec
    inet6 fd00::6726:a4fe:5a09:e320/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86359sec preferred_lft 14359sec
    inet6 fe80::dfd0:9539:db96:5bad/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ed:31:01 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.11/24 brd 192.168.56.255 scope global noprefixroute enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::2305:db34:a05:7660/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
mint@mint:~$
```

```
mint@mint:~$ lsb_release -a
No LSB modules are available.
Distributor ID: LinuxMint
Description:    Linux Mint 22.2
Release:        22.2
Codename:       zara
```